

Equilibrium Price & Quantity

LET – Mathematics and Statistics

No taxation

1. Given the following functions for demand and for supply, find the equilibrium price and quantity.
 - (a) Demand: $P = 65 - Q$, supply: $P = 11 + Q$
 - (b) Demand: $4P = 100 - 3Q$, supply: $P = 2Q + 10$
 - (c) Demand: $P = -Q^2 - Q + 88$, supply: $P = Q + 8$
 - (d) Demand: $4P = -Q^2 - 2Q + 100$, supply: $P = 2Q + 1$

Taxation

2. Demand for an item is governed by $P = 100 - Q$, with supply given by $P = Q + 20$. Price is measured in pounds and quantity in 1000 units.
 - (a) Given no tax, what is the equilibrium price and quantity?
 - (b) If tax is set at £0.30 per unit, what is the equilibrium price and quantity, and what is the tax revenue?
 - (c) What is the equilibrium price and quantity if tax is set at 5%
 - (d) Recalculate the equilibrium price and quantity, for parts a, b, and c, if demand is now given by

$$P = 100 - 0.5Q - 0.1Q^2$$